



NutriTrusto

Your Smart Pantry Companion

The Impact of Real-Time Nutritional Transparency and Smart Pantry Management on Food Waste Reduction and Health Awareness Among Urban Consumers

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PROJECT DETAILS

Branch: CSE

Guide: Guide's Name

Year: H&M 2026

Phase: Empathize & Define

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Abstract

NutriTrusto is a mobile-first web application addressing two interconnected problems faced by urban consumers across India: inability to evaluate nutritional quality of packaged food at point of purchase, and lack of a system to manage perishable grocery inventory. By combining real-time barcode-based health ratings with an AI-powered smart pantry tracker, NutriTrusto enables informed food choices while reducing household food waste.

Problem Space

Health + Waste

Target Users

All Urban Consumers

Survey Size

79 Respondents

Phase

Empathize & Define

Barcode Health Scanner

Scan barcode -> instant NutriTrust score, ingredient risk flags, diet conflict alerts, globally banned ingredient checks.

Smart Pantry Manager

Track groceries, monitor expiry countdowns, upload receipts, and get AI recipes before items spoil.

Motivation & Why Us?

Motivation to Solve This Problem

Living the problem

As urban consumers ourselves, we experienced buying packaged food with limited knowledge, forgetting perishables, and struggling to decode additive names.

Market gap identified

FactsScan and TruthIn stop at scan-level interpretation. No Indian app combines barcode health rating and smart pantry management together.

Real social impact

Across households and occupations, consumers need tools to make informed food choices and reduce avoidable grocery waste.

Why Us?

1

Domain proximity

We face this problem daily as urban consumers managing our own kitchens and groceries.

2

Technical capability

Built with Next.js, Supabase, Gemini AI and Open Food Facts; deployed on Vercel.

3

Unique insight

Barcode scanning and pantry management are two sides of one behavior loop.

4

Validation-first

79-respondent survey completed before build; decisions are evidence-driven.

User Research & Ideal Customer Profile

IDEAL CUSTOMER PROFILE	
Who	Urban consumer 18-60 in Tier 1 and Tier 2 Indian cities
Location	Bengaluru, Mumbai, Delhi, Hyderabad, Pune, Chennai
Behaviour	Buys packaged food regularly, uses delivery apps, no pantry system
Occupation	Students, professionals, homemakers, self-employed, retired
Pain 1	Cannot decode E-numbers/additives; avg label understanding 2.9/5
Pain 2	Groceries expire before use; 73% waste food regularly
Goal	Make informed choices and reduce waste with minimal effort

Research Methods Used

Empathy Interviews	Qualitative
1:1 conversations with 8 urban consumers across occupations	
Observational Study	Qualitative
Observed grocery behavior and label-reading patterns	
Online Survey	Quantitative
30-question form; 79 respondents across 6 occupation groups	
Competitive Analysis	Secondary
FactsScan, TruthIn, Yuka, KitchenPal gap mapping	

ICP Validation - Survey First-Cut Results (n=79)

47%

consume packaged food weekly or daily

94%

have NO structured pantry tracking system

71%

scored 4 or 5 likelihood to use NutriTrusto

84%

find labels confusing (Q9)

Top 8 Features Wanted by Respondents (Q29)



Survey validates demand across all demographics - 71% expressed high intent to use NutriTrusto.

Empathy Artifacts - Empathy Map

Empathy Map - Urban Consumer (Packaged Food User)

THINKS & FEELS

- Should eat healthier but labels are too complex
- Feels guilty wasting food but forgets purchases
- Unsure whether common snacks are harmful

HEARS

- That ingredient is banned abroad
- You should track food but no easy tool exists
- Family/roommate complaints about spoiled groceries

SAYS & DOES

- Buys familiar brands without checking labels
- Forgets perishables until late-stage spoilage
- Orders delivery when pantry appears empty

PAIN POINTS

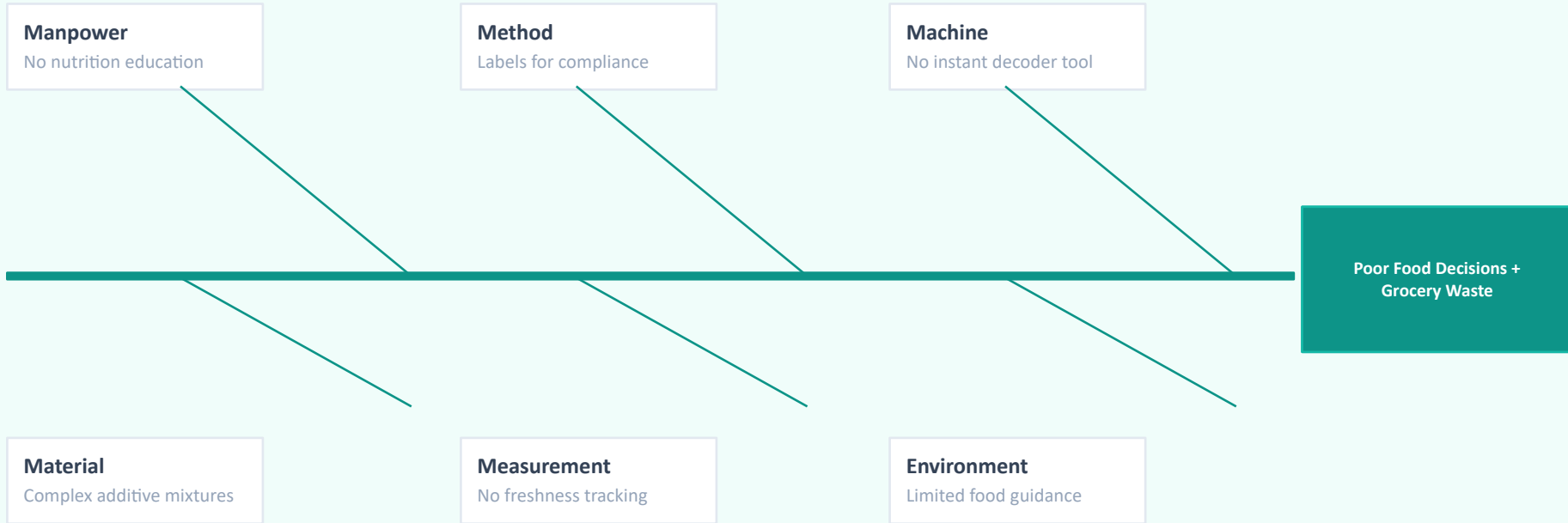
- 84% find labels confusing (Q9)
- 73% waste food regularly (Q23)
- 94% have no structured pantry system (Q19)

Problem Discovery & Definition - Methods Used



Root Cause Analysis - Ishikawa Fishbone Diagram

Why do urban consumers make uninformed food choices and waste groceries?



5 Whys Technique - Root Cause Drill-Down

The 5 Whys technique (HUM 4412 Module 5) asks why repeatedly to uncover root cause.

Chain 1 - Nutritional Ignorance		Chain 2 - Food Waste	
1	Consumers buy unhealthy food without knowing ingredient risks	1	Consumers waste groceries regularly (73% at least sometimes)
2	They do not check ingredient labels before purchase	2	They forget pantry items and expiry timelines
3	Labels are complex (2.9/5 understanding, 84% confusion)	3	94% have no tracking system (Q19)
4	No simple real-time decoding tool exists	4	No pantry app for everyday Indian household behavior
5	ROOT CAUSE: No accessible food intelligence tool for Indian consumers	5	ROOT CAUSE: No purchase-to-pantry expiry-aware management app

Both root causes are solved by a single solution - NutriTrusto

Survey Findings - Nutritional Awareness (n=79)

33%

rarely or never check ingredients
(Q4 scores 1-2)

84%

find ingredient labels moderately
to very confusing (Q9)

2.9/5

average label comprehension
score (Q8)

80%

compare brands before buying
(Q6)

How often do you check ingredients? (Q4 - scale 1 to 5)

Score 1 - Never



13%

Score 2



20%

Score 3 - Sometimes



41%

Score 4



18%

Score 5 - Always



9%

Key insight: Despite 76% having some awareness of additive risks, 73% do not consistently check labels. The barrier is label complexity, not awareness.

Survey Findings - Food Waste & Expiry (n=79)

39%

consumed expired food
(unknowingly or knowingly)

73%

waste food at least
sometimes (Q23)

94%

no structured pantry
tracking system (Q19)

4.0/5

expiry reminders rated
helpful (Q21 avg)

How respondents currently track groceries (Q19)

Try to remember (56%)



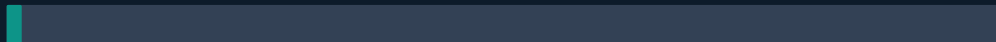
Just check the fridge (38%)



Write it down (5%)



Use an app (1%)



Food waste frequency (Q23)

Never 11%



Rarely 15%



Sometimes 35%



Often 32%



Very often 6%



94% have no pantry system - the issue is not willingness, but absence of a simple, accessible tool.

User Personas - Derived from Survey & Research

Based on HUM 4412 Customer Persona framework - research-backed profiles of target users.

PERSONA 1 - THE ACTIVE CONSUMER

RP**Rahul P.**

Age 20 · Bengaluru · Undergrad Student

- Buys packaged food daily
- Rarely checks labels (score 1-2)
- Tracks pantry by visual fridge checks
- Goal: quick, low-effort healthy decisions
- Pain: additive names are unclear

"I just grab what I know. I need instant ingredient clarity."

PERSONA 2 - THE MINDFUL HOMEMAKER

SK**Sujata K.**

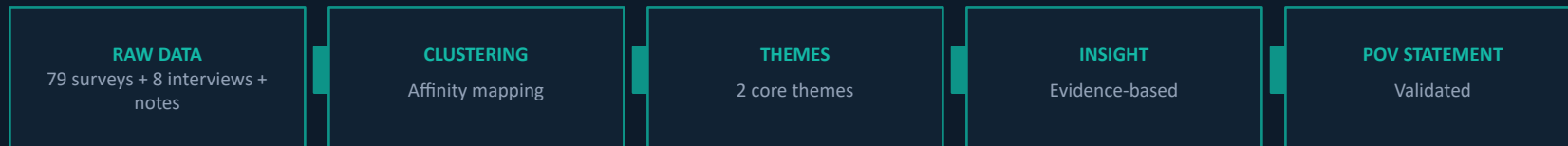
Age 38 · Mumbai · Homemaker

- Buys packaged products occasionally
- Checks labels sometimes (score 3)
- Tracks groceries by memory/list notes
- Goal: reduce monthly household waste
- Pain: expiry and label complexity

"I want better decisions, but labels are hard to decode quickly."

From Data to Robust Insights

Problem definition journey - raw data to validated POV



Theme 1: Nutritional Blindness at Purchase

73% do not consistently check labels; 84% report confusion.

94% no pantry system -> reactive, not proactive behavior

39% consumed expired food -> present safety risk

Theme 2: Reactive Pantry Management

94% rely on memory/fridge checks; 73% waste food regularly.

33% never check labels -> complexity barrier

71% high intent to use NutriTrusto -> validated demand

Insights directly shaped the final POV statement and NutriTrusto core features

Final Problem Definition

Point of View (POV) Statement

Urban consumers need a way to make informed food choices and avoid waste - because they shop with limited nutritional knowledge and have no system to track what they own or when it expires.

Validated Problem Statement

Urban consumers across India lack accessible tools to evaluate packaged-food nutrition at point of purchase and systematically track perishable groceries, resulting in uninformed dietary choices and avoidable waste.

Contributing Sub-Problems (Survey-Validated)

P1	No real-time tool to decode Indian packaged food labels at purchase	Q4,Q8,Q9
P2	No integrated system connecting purchase to expiry-aware pantry tracking	Q19,Q21
P3	Low awareness of additives, processing levels, and banned ingredients	Q11,Q13
P4	No AI recipe guidance to utilize food before spoilage	Q23,Q26

Research Objectives & Hypotheses

Research Objectives

1

O1: Assess impact of real-time ratings on informed purchase behavior

2

O2: Evaluate smart pantry tracking + expiry alerts in reducing waste

3

O3: Examine AI recipe suggestion effect on ingredient utilization

4

O4: Measure acceptance of NutriTrusto using TAM

Research Hypotheses

H1

Real-time barcode info -> informed healthier purchase decisions

HBM

H2

Expiry alerts -> significantly reduced food wastage

Nudge

H3

AI recipes from expiring ingredients -> higher utilization

Nudge

H4

Diet-preference filtering -> improved trust and engagement

TAM

H5

Visual ratings outperform raw labels in influencing choice

HBM+TAM

Research Methodology & Questionnaire Design

Research Design Descriptive + Exploratory mixed-method design	Research Approach Quantitative survey + Qualitative interviews & observation
Target Population Urban consumers 18-60 across all occupation types	Sampling Method Purposive + Snowball via networks and communities
Sample Size 79 respondents achieved (target: 100+)	Data Collection 30-Q form + interviews + observational study
Data Analysis Descriptive stats, Likert scale, thematic coding, affinity mapping	Tech Stack Next.js, Supabase, Open Food Facts API, Gemini AI, Vercel

Survey Structure - 30 Questions across 4 Stages

Stage 1	Stage 2	Stage 3	Stage 4
Food Habits · Q1-Q9	Safety & Expiry · Q10-Q22	Waste Behaviour · Q23-Q26	App Intent · Q27-Q30